

Lab Report: 04

Experiment Name: Configuration and Verification of Static Routing in a Three-Router Topology

1. Introduction

Static routing is a method where a network administrator manually configures the routing table of a router to define specific paths to destination networks. Unlike dynamic routing, static routes do not update automatically if the topology changes. This lab focuses on establishing end-to-end connectivity between three separate Local Area Networks (LANs) connected via three routers in a serial chain.

2. Tools

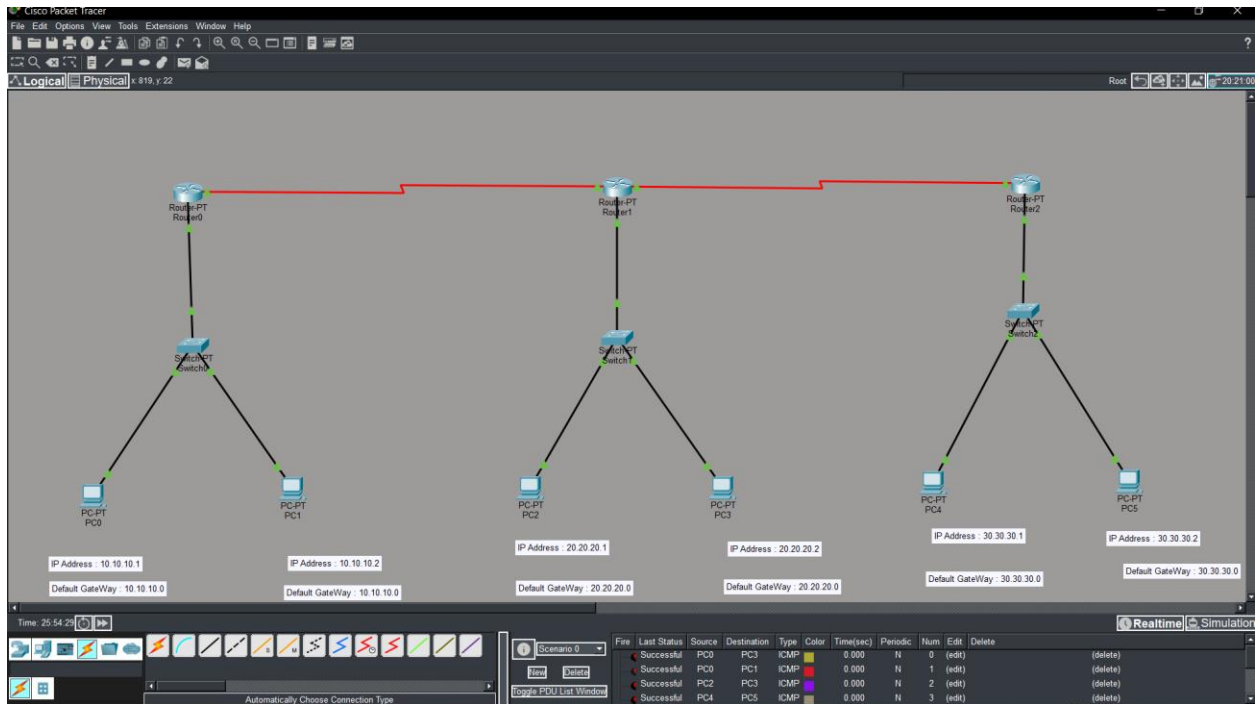
- Simulation Software: Cisco Packet Tracer.
- Network Devices: 3x Router-PT, 3x Switch-PT.
- End Devices: 6x Generic PCs (PC0 – PC5).
- Cables: Serial DCE (Router-to-Router) and Copper Straight-Through (Router-to-Switch/PC).

3. Procedure

1. **Topology Setup:** Drag and drop three routers, three switches, and six PCs into the workspace. Connect them according to the diagram.
 2. **IP Addressing:** * Assign IP addresses to each PC according to their respective subnets (10.10.10.x, 20.20.20.x, and 30.30.30.x).
 - Configure the FastEthernet and Serial interfaces on each router.
 3. **Static Route Configuration:** On each router, manually enter the ip route command for any network that is not directly connected.
 - *Example Command:* Router(config)# ip route [remote_network] [mask] [next_hop_address]
 4. **Connectivity Test:** Use the "Simple PDU" tool to send packets between PCs in different networks.
 5. **Verification:** Use the show ip route command in the CLI to verify that the static routes are active.
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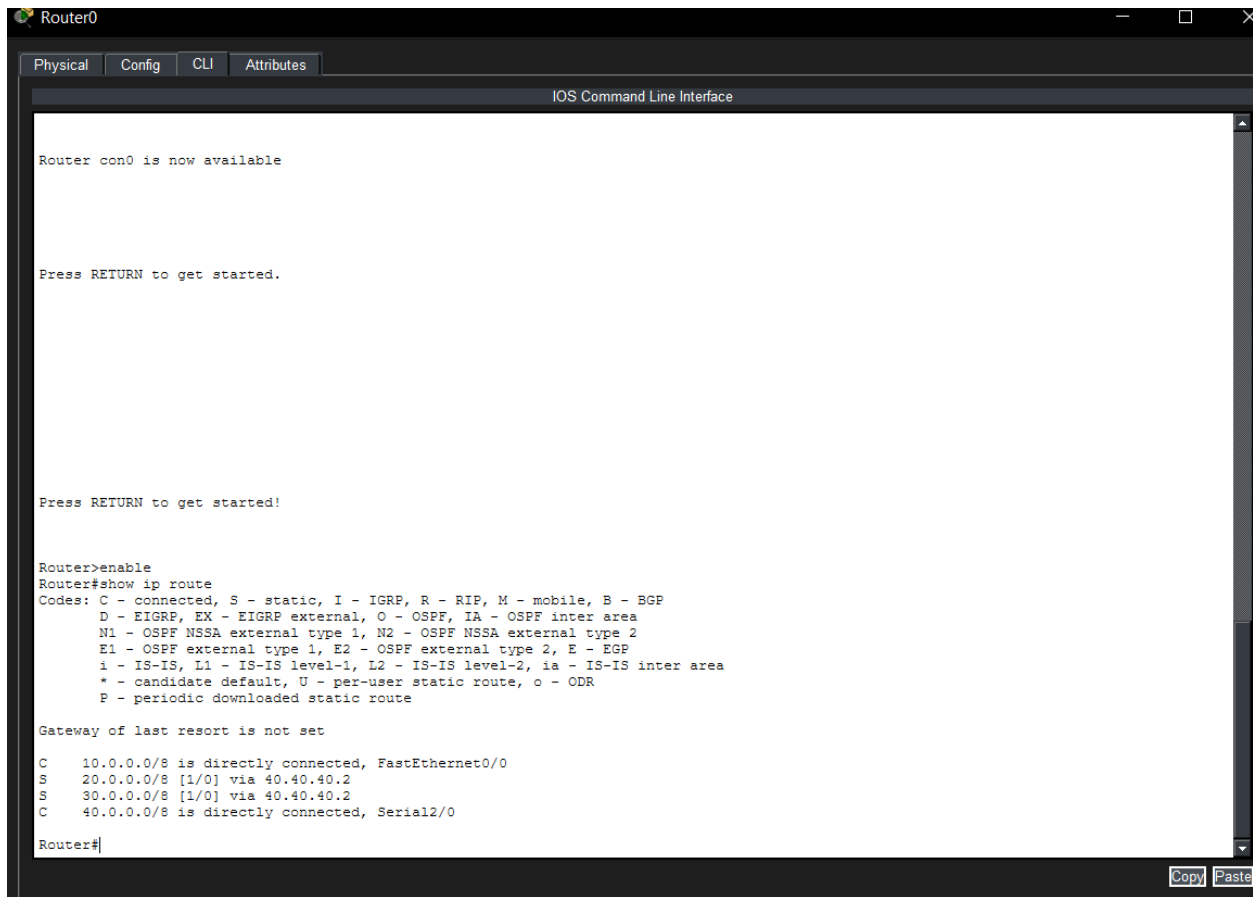
4. Results

A. Network Topology



Description: Figure 1: The completed network diagram showing all green link lights.

B. Routing Table Verification (CLI)



The screenshot shows the Router0 CLI interface with the following text:

```
Router con0 is now available

Press RETURN to get started.

Press RETURN to get started!

Router>enable
Router#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

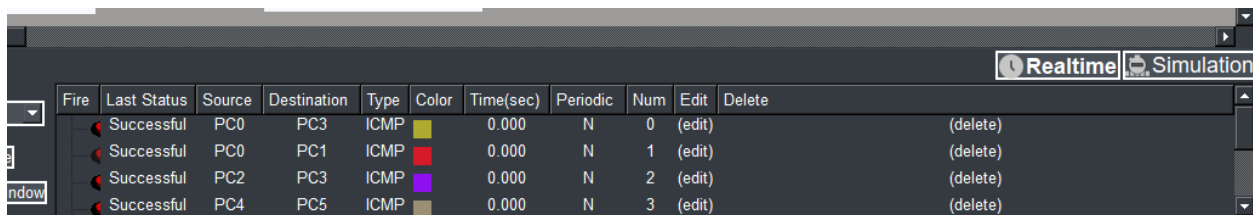
Gateway of last resort is not set

C    10.0.0.0/8 is directly connected, FastEthernet0/0
S    20.0.0.0/8 [1/0] via 40.40.40.2
S    30.0.0.0/8 [1/0] via 40.40.40.2
C    40.0.0.0/8 is directly connected, Serial2/0

Router#
```

Description: Figure 2: The routing table displaying static routes (marked with 'S') to remote networks.

C. Connectivity Success (PDU List)



The screenshot shows the Packet Tracer PDU list with the following data:

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC0	PC3	ICMP	Yellow	0.000	N	0	(edit)	(delete)
	Successful	PC0	PC1	ICMP	Red	0.000	N	1	(edit)	(delete)
	Successful	PC2	PC3	ICMP	Purple	0.000	N	2	(edit)	(delete)
	Successful	PC4	PC5	ICMP	Brown	0.000	N	3	(edit)	(delete)

Description: Figure 3: Packet Tracer PDU list confirming successful ICMP communication between PC0, PC3, and PC5.

5. Conclusion

In this experiment, we successfully implemented static routing across a three-router topology. By manually defining the next-hop addresses, we allowed data to travel across multiple router hops.

Key Observations:

- **Static routing provides a high level of security because the administrator has full control over the paths.**
- **It is efficient for small networks as it does not produce routing protocol traffic (overhead).**
- **However, if a link fails, the network remains down until the administrator manually re-routes the traffic, which is a major limitation compared to dynamic routing.**